your result to the World Geographic Society.) Since the singularity is so small, 10^{-33} centimeter, and is at the precise center of the hole, the distance from singularity to horizon should be equal to the horizon's radius. You are tempted to calculate this radius by the standard method of dividing the circumference by 2π (6.283185307 . . .). However, in your studies on Earth you were warned not to believe such a calculation. The hole's enormous gravitational pull completely distorts the geometry of space inside and near the hole,[17] in much the same manner as an extremely heavy rock, placed on a sheet of rubber, distorts the sheet's geometry (Figure P.3), and as a result the horizon's radius is not equal to its circumference divided by 2π.

"Never mind," you say to yourself. "Lobachevsky, Riemann, and other great mathematicians have taught us how to calculate the properties of circles when space is curved, and Einstein has incorporated those calculations into his general relativistic description of the laws of gravity. I can use these curved-space formulas to compute the horizon's radius."

But then you remember from your studies on Earth that, although a black hole's mass and angular momentum determine all the properties of the hole's horizon and exterior, they do not determine its interior. General relativity insists that the interior, near the singularity, should be chaotic and violently nonspherical,[18] much like the tip of the rubber sheet in Figure P.3 if the heavy rock in it is jagged and is bouncing up and down wildly. Moreover, the chaotic nature of the hole's core will depend not only on the hole's mass and angular momentum, but also on the details of the stellar implosion by which the hole was born, and the details of the subsequent infall of interstellar gas—details that you do not know.

"So what," you say to yourself. "Whatever may be its structure, the chaotic core must have a circumference far smaller than a centimeter. Thus, I will make only a tiny error if I ignore it when computing the horizon's radius."

But then you remember that space can be so extremely warped near the singularity that the chaotic region might be millions of kilometers in radius though only a fraction of a centimeter in circumference, just as the rock in Figure P.3, if heavy enough, can drive the chaotic tip of the rubber sheet exceedingly far downward while leaving the circumference of the chaotic region extremely small. The errors in your calculated radius could thus be